Allen-Bradley Guardmaster®

MINOTAUR MSR127R-MSR127T

(a) **MONITORING SAFETY RELAY**
ÜBERWACHUNGS-/SICHERHEITSRELAIS
RELAIS DE SECURITE POUR CONTROLE

(b) Installation Instructions

RETAIN THESE INSTRUCTIONS

Installation must be in accordance with the following steps and must be carried out by suitably competent personnel.
This device is intended to be part of the safety related control system of a machine. Before installation, a risk assessment should be performed to determine whether the specifications of this device are suitable for all foreseeable operational and environmental characteristics of the machine to which it is to be fitted.
At regular intervals during the life of the machine check whether the characteristics foreseen remain valid.
Guardmaster cannot accept responsibility for a failure of this device if the procedures given in this sheet are not implemented or if it is used outside the recommended specifications in this sheet.
Exposure to shock and/or vibration in excess of those stated in IEC 68 part: 2-6/7 should be prevented.
Adherence to the recommended inspection and maintenance instructions forms part of the warranty.

Einbauanleitung

DIESE ANLEITUNG AUFBEWAHREN

Die Installation muß unter Einhaltung der nachstehend beschriebenen Schritte, und durch geeignetes, fachlich qualifiziertes Personal erfolgen.
Diese Vorrichtung ist als Teil des sicherheitsrelevanten Kontrollsystems einer Maschine vorgesehen. Vor der Installation sollte eine Risikobewertung zur Festlegung dessen erfolgen, ob die Spezifikationen dieser Vorrichtung für alle vorhersehbaren betrieblichen und umweltbezogenen Eigenschaften der jeweiligen Maschine geeignet sind, an der sie installiert werden soll.
In regelmäßigen Abständen während der Lebensdauer der Maschine ist zu überprüfen, ob die vorhergesehenen Eigenschaften weiterhin gültig sind. Guardmaster kann keinerlei Verantwortung für ein Versagen dieser Vorrichtung übernehmen, wenn die in diesem Schriftblatt gegebenen Verfahrensweisen nicht implementiert wurden, oder wenn sie außerhalb der auf diesem Schriftblatt empfohlenen Spezifikationen verwendet wird.
Eine Aussetzung an Stoßbelastungen und/oder Vibrationen, die überhalb den in IEC 68, Teil 2-6/7 angegebenen Werten liegen, sollte verhindert werden.
Die Einhaltung der empfohlenen Inspektions- und Wartungsvorschriften ist Teil der Garantie.

Notice D'installation

CONSERVEZ CES INSTRUCTIONS

L'installation doit être effectuée conformément aux instructions suivantes, par des membres qualifiés du personnel.
Ce dispositif est étudié pour être incorporé dans le système de contrôle pour la sécurité d'une machine. Avant l'installation, on doit effectuer une évaluation des risques pour déterminer si les spécifications de ce dispositif sont appropriées pour toutes les caractéristiques de service et du milieu d'utilisation prévues pour la machine sur laquelle il sera monté.
Vérifier, à des échéances régulières au cours de la vie de la machine, que les caractéristiques prévues sont toujours valables. Guardmaster décline toute responsabilité pour les défaillances de cet appareil si les procédures décrites dans la présente notice ne sont pas appliquées ou si l'appareil est utilisé hors des spécifications recommandées dans cette même notice.
Eviter toute exposition à des chocs et/ou des vibrations supérieurs à ceux qui sont spécifiés dans la norme IEC 68 part. 1-6/7.
Le respect des instructions relatives à l'inspection, au contrôle et à l'entretien de cet appareil rentre dans l'application de la garantie.

(c) Mode of Operation

The dual-channel operation shown in wiring examples 5 and 6 includes crossfault monitoring between both E-stop circuits. That means in case of shorts between the two E-stop channels the **MSR127** will de-energise the outputs. This is achieved by an electronic protection circuit in the safety relay. After elimination of the malfunction, the **MSR127** is ready for operation again.
MSR 127R VERSION
The versions with monitored start supervise the start circuit and will only activate the **MSR127** if contacts via terminals S12 and S34 are closing during start conditions.
MSR127T VERSION
Models with autostart function will be activated automatically by the supply voltage, if the E-stop circuits and the feedback loop are closed.
A start-push-button may be integrated between the feedback-loop (S12-S34) for manual reset.
In autostart-applications where both circuits are not closed simultaneously, (e.g. safety gates) channel 2 has to be activated before channel 1.
If the inputs of the **MSR127** are activated with external 24VDC, the negative pole has to be connected to S21 (e.g. Light curtain applications). In those applications power supply on A1-A2 is only necessary to drive the Power-LED.
To control NC contacts from external contactors the feed-back loop should be connected in series between S12 and S34.
The relay is available with fixed or removable terminals.

Funktionsweise

Bei 2-kanaliger Ansteuerung gemäß Schaltungsbeispielen 5 und 6 besteht Querschlußsicherheit. Das heißt, bei einem Leitungsschluss spricht eine elektronische Sicherung im Gerät an und schaltet das **MSR127RTP** aus.
Nach Beseitigung des Fehlers ist das **MSR127RTP** wieder betriebsbereit.
Bei Geräten mit überwachtem Start wird der Starttaster bei jedem Einschaltvorgang überprüft. Ist der Eintaster vor dem Entriegeln der Not-Aus-Taster oder Anlegen der Versorgungsspannung geschlossen, ist kein Start möglich.
Geräte mit Autostartfunktion schalten automatisch bei anliegender Versorgungsspannung ein, sofern die Not-Aus Kreise und der Rückführkreis geschlossen sind. Ein Starttaster kann auch hier in den Rückführkreis (S12-S34) eingebunden werden (manueller Start).
Werden bei Autostart-Anwendungen die Not-Aus Kreise nicht gleichzeitig betätigt, so muß Kanal 2 vor Kanal 1 geschlossen werden (z. B. Schutztürüberwachung).
Werden die Eingänge des **MSR127RTP**, beispielsweise durch ein Sicherheitslichtgitter, extern mit 24VDC angesteuert, so ist das negative Potential mit S21 zu verbinden.
Zu überwachende Öffnerkontakte von externen Erweiterungen sind zwischen S12 und S34 bzw. in Reihe mit dem Starttaster zu schalten.
Die Geräte sind wahlweise mit festen oder abnehmbaren Klemmenblöcken erhältlich.

Mode de Fonctionnement

La configuration bi-canal illustrée dans les exemples de câblage 3 et 5 prévoit le contrôle des défaillances entre les deux circuits d'arrêt d'urgence. En cas de court-circuit entre les deux canaux d'arrêt d'urgence, le MSR127RTP désactive donc les sorties grâce au circuit de protection électronique prévu dans le relais de sécurité.
Après élimination du défaut, le MSR127RTP est de nouveau prêt à fonctionner.
Les versions à initialisation contrôlée commandent le circuit de mise en route et activent uniquement le MSR127RTP si les contacts des bornes S12 et S14 se ferment pendant l'initialisation.
Les modèles à fonction d'autoinitialisation sont automatiquement activés par la tension d'alimentation si les circuits d'arrêt d'urgence et la boucle de retour sont fermés.
Il est possible de monter un bouton de lancement entre la boucle de feed-back (S12-S34) pour prévoir l'initialisation manuelle.
Pour les applications à initialisation automatique dans lesquelles les deux circuits se ferment simultanément (par exemple les portes de sécurité), le canal 2 doit être activé avant le canal 1.
Si les entrées du MSR127RTP sont activées par une alimentation externe de 24 V c.c., le pôle négatif doit être connecté à S21 (par ex. pour les barrières photoélectriques). Dans ces applications, l'alimentation électrique de A1-A2 est uniquement sollicitée pour alimenter le voyant de marche.
Pour commander les contacts N/F à partir des contacteurs externes, connecter la boucle de retour en série entre S12 et S34.
Le relais est disponible avec bornes fixes ou amovibles.

Deutsch / Français

1

(a) **Rückansicht** / *Vue de l'arrière*
(b) **Spannung abschalten/** *Isoler les alimentations*
(c) **Auf 35mm-Normschiene anbringen /** *Montage sur rail DIN 35mm*
(d) **In Einbaugehäuse nach mind. IP 54 montieren /** *Monter dans un coffret conforme au minimum à la norme IP 54*

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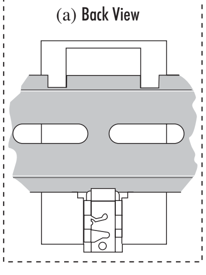
(b)
A1 & A2 = Spannungsversorgung
S52 & S12 = Schutzeingang (Ruhekontakt)
S21 & S22 = Schutzeingang (Ruhekontakt)
S11 & S52 = Schutzeingang (einkanalig) (Ruhekontakt)
S12 & S34 = Überwachungsrückmeldungschleife einschliesslich Rückstellung-Taster
13 & 14 = Schutzausgang 1 (Arbeitskontakt)
23 & 24 = Schutzausgang 2 (Arbeitskontakt)
33 & 34 = Schutzausgang 3 (Arbeitskontakt)
41 & 42 = Hilfsausgang (Ruhekontakt) /
A1 & A2 = Alimentation
S52 & S12 = Entrée de sécurité (N/F)
S21 & S22 = Entrée de sécurité (N/F)
S11 & S52 = Entrée de sécurité (monocanal) N/F)
S12 & S34 = Boucle de retour de contrôle avec bouton d'initialisation incorporé
13 & 14 = Sortie de sécurité 1 (N/O)
23 & 24 = Sortie de sécurité 2 (N/O)
33 & 34 = Sortie de sécurité 3 (N/O)
41 & 42 = Sortie auxiliaire (N/F)
41 & 42 = sortie auxiliaire 1 (N/F)

1

(a) Back View

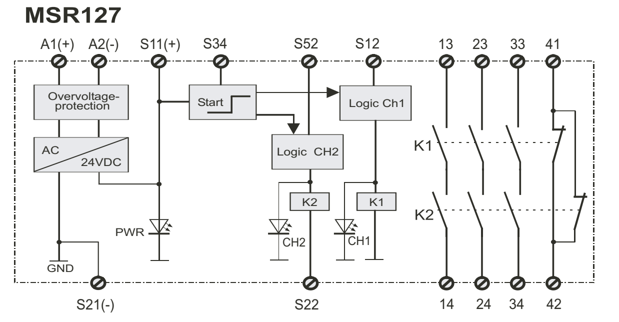


(b) Isolate power

(c) Mount on 35mm DIN rail.
(d) Mount in enclosure to a min. of IP 54.

(c) **LED Anzeigen:**
Strom (GRÜN) - Leuchtet auf, wenn Strom eingeschaltet ist
CH1 (GRÜN) - Leuchtet auf, wenn K1 geschlossen ist
CH2 (GRÜN) - Leuchtet auf, wenn K2 geschlossen ist /
Voyants:
Alimentation (VERTE) - allumée à la mise sous tension
CH1 (VERTE) - allumée lorsque K1 est fermé
CH2 (VERTE) - allumée lorsque K2 est fermé

(d) CIRCUIT DIAGRAM/AUSCHLUSSDIAGRAMM/SCHEMA DES CONNEXIONS



2 (a) **Connections / Anschlüsse / Connexions**

(b)
A1 & A2 = Power
S52 & S12 = Safety input (N/C)
S21 & S22 = Safety input (N/C)
S11 & S52 = Safety input (single channel) (N/C)
S12 & S34 = Monitoring feedback loop Incorporating reset button
13 & 14 = Safety output 1 (N/O)
23 & 24 = Safety output 2 (N/O)
33 & 34 = Safety output 3 (N/O)
41 & 42 = Auxilary output (N/C)

13	23	33	41
A1	S11	S52	S12

(c) **LED Indication**

- **PWR (GREEN)** - Illuminates when power on
- **CH1 (GREEN)** - Illuminates when K1 is closed
- **CH2 (GREEN)** - Illuminates when K2 is closed

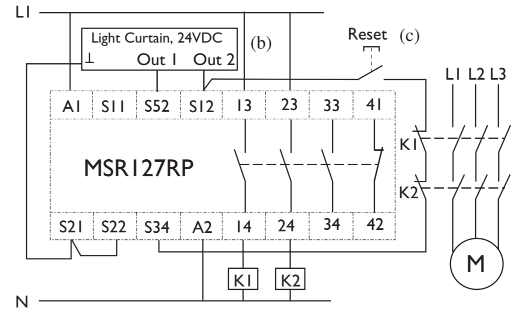
S21	S22	S34	A2
14	24	34	42

Deutsch / Français

3

(b) Lichtschranke
Ausgang 1 Ausgang 2 /
Barrière photoélectrique
Out 1 Out 2
(c) Rückstellung / *Initialisation*
(d) Lichtschranke, manuelle Rückstellung, überwachter Ausgang /
Barrière photoélectrique, initialisation manuelle, sortie non contrôlée

3 (a) Wiring example 1 / Anschlußbeispiele 1 / Exemples de câblage 1



(d) **Light Curtain, Manual Reset, Monitored Output**

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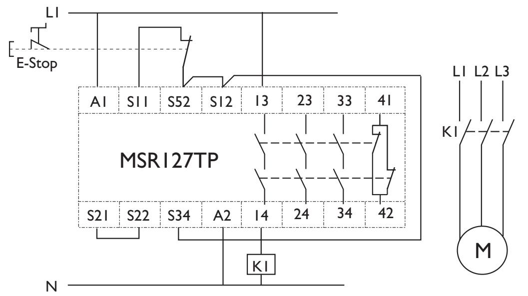
(b) Einkanal-Notaus, automatische Rückstellung, keine Ausgangsüberwachung / *A.d'urgence monocanal, autoinitialisation, sortie non contrôlée*

4 (a) Wiring example 2 / Anschlußbeispiele 2 / Exemples de câblage 2

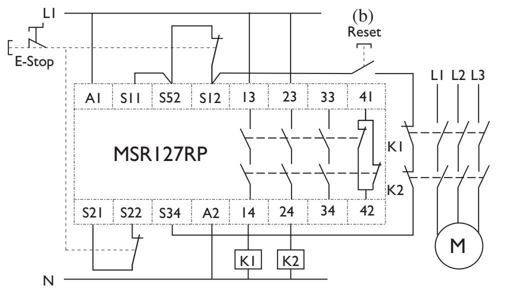


(b) **Single Channel E-Stop, Auto. Reset, No Output Monitoring**

5

(b) Rückstellung / *Initialisation*
(c) Zweikanal-Notaus, manuelle Rückstellung, überwachter Ausgang/
A.d'urgence bi-canal, initialisation manuelle, sortie contrôlée

5 (a) Wiring example 3 / Anschlußbeispiele 3 / Exemples de câblage 3



(c) **Dual Channel E-Stop, Manual Reset, Monitored Output**

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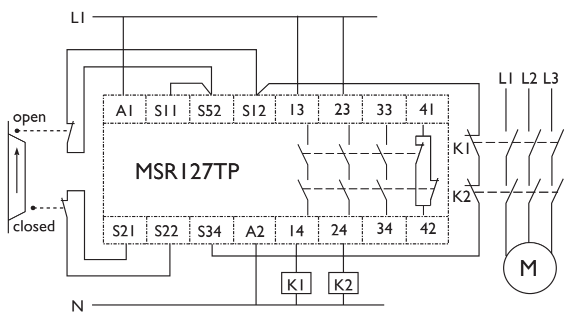
(b) Zweikanal-Sicherheitstore, automatische Rückstellung, überwachter Ausgang /
Portes de sécurité bi-canal, autoinitialisation, sortie contrôlée

6 (a) Wiring example 4 / Anschlußbeispiele 4 / Exemples de câblage 4

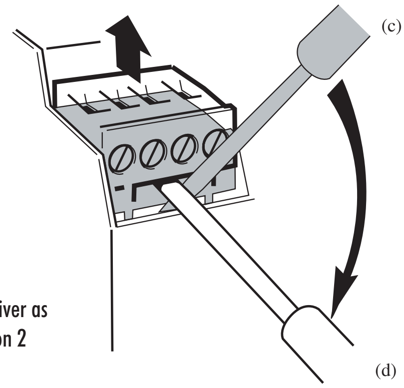


(b) **Dual Channel Safety Gates, Auto. Reset, Monitored Output**

7

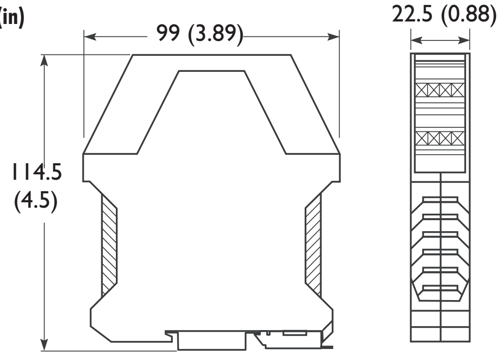
(a) Abnehmbare Klemmen - nur bei 'P'-Ausführungen / *Bornes amovibles — disponibles sur versions P uniquement*
(b) Um die Klemmen abzunehmen - Schraubenzieher in Position 1 ansetzen und langsam in Position 2 bringen/ *Pour ôter les bornes : insérer un tournevis au repère 1 et baisser lentement jusqu'au repère 2*
(c) Position 1 / *Repère 1*
(d) Position 2 / *Repère 2*

7



(e) DIMENSION DIAGRAM/ABMESSUNGSDIAGRAMM/SCHEMA COTÉ

Dimensions - mm (in)



(f) Technical Specifications

Standards	IEC/EN60204-1, ISOTR12100, ISO13849-1(EN954-1)
Safety Category	Cat. 4 per EN 954-1
Approvals	CE marked for all applicable directives, cULus and BG
Power Supply	24V AC/DC, 115V AC or 230V AC 0.85 to 1.1 x rated voltage, 50/60 Hz
Power Consumption	2W
Safety Inputs	1 N.C. or 2 N.C. or 2 PNP light curtain
Input Simultaneity	Infinite
Max. Allowable Input Resistance	110 ohms
Reset	Monitored Manual or Auto./Manual
Outputs	3 N.O. Safety; 1N.C. Auxiliary
Output Rating B300 / P300 ref.: I_{th}, U_i	B300, AC-15, 5A/250V AC P300, DC-13, 3A/24V DC
Fuses Output (external)	6A Slow Blow or 10A Quick Blow
Min. Switched Current/Voltage	10mA/10V
Contact Material	$AgSnO_2 + 0.5\mu Au$
Power On Delay	1s
Response Time	15ms
Recovery Time	100ms
Impulse Withstand Voltage	2500V
Pollution Degree	2
Operating Temperature	-5 °C to +55 °C (+23 °F to 131 °F)
Humidity	90% RH
Enclosure Protection	IP40 (NEMA 1)
Terminal Protection	IP20
Wiring:	Use copper that will withstand 60/75 °C
Conductor Size	0.2-4mm² (24-12AWG)Wire Size Only
Torque Settings - terminal screws	0.6 Nm - 0.8Nm (5 - 7 lb•in)
Case Material	Polyamide PA 6.6
Mounting	35mm DIN rail
Weight 24V DC 110V AC or 230V AC	210g (0.463lbs) 260g (0.573lbs)
Electrical Life 220V AC/4A/880VA cosφ=0.35 220V AC/1.7A/375VA cosφ=0.6 30V DC/2A/60W 10V DC/0.01A/0.1W	 100,000 operations 500,000 operations 1,000,000 operations 2,000,000 operations
Mechanical Life	2,000,000 cycles
Vibration	10-55 Hz, 0.35mm

Technische Daten

Normen	IEC/EN60204-1, ISOTR12100, ISO13849-1(EN954-1)
Schutzkategorie	Kat. 4 gem. EN 954-1
Zulassungen	CE-Kennzeichnung für alle zutreffenden Direktiven, cULus und BG
Stromversorgung	24V AC/DC, 115V AC oder 230V AC 0,85 bis 1,1 x Nennspannung, 50/60 Hz
Leistungsverbrauch	2W
Schutzeingänge	1 Ruhekontakt oder 2 Ruhekontakt oder 2 PNP Lichtschranken
Eingangsgleichzeitigkeit	Unbegrenzt
Max. zulässiger Eingangswiderstand	110 ohms
Rückstellung	überwachte manuelle oder automatische/manuelle
Ausgänge	3 Arbeitskontakte-Schutzausgang; 1 Ruhekontakt-Hilfsausgang
Ausgangsnennbelastung B300 / P300 ref.: I_{th}, U_i	B300, AC-15, 5A/250V AC P300, DC-13, 3A/24V DC
Sicherungen Ausgang (extern)	6A träge oder 10A flinke
Min. geschalteter Strom/Spannung	10mA/10V
Kontaktmaterial	$AgSnO_2 + 0.5\mu Au$
Strom-ein-Verzögerung	1s
Reaktionszeit	15ms
Erholungszeit	100ms
Stehstossspannung	2500V
Verschmutzungsgrad	2
Betriebstemperatur	-5 °C bis +55 °C
Feuchtigkeit	90% RH
Gehäuseschutz	IP40 (NEMA 1)
Klemmenschutz	IP20
Leitungsmaterial:	Kupferdraht mit Temperatur- beständigkeit von 60/75 °C.
Leiterquerschnitt	0.2-4mm² (24-12AWG)
Drehmomentwerte - Klemmenschrauben	0.6 Nm - 0.8Nm (5 - 7 lb•in)
Gehäusematerial	Polyamid PA 6.6
Befestigung	35mm DIN-Schiene
Gewicht 24V DC 110V AC oder 230V AC	210g (0.463lbs) 260g (0.573lbs)
Elektrische Lebensdauer 220V AC/4A/880VA cosφ=0.35 220V AC/1.7A/375VA cosφ=0.6 30V DC/2A/60W 10V DC/0.01A/0.1W	 100,000 Betätigungen 500,000 Betätigungen 1,000,000 Betätigungen 2,000,000 Betätigungen
Mechanische Lebensdauer	2,000,000 Arbeitstakte
Vibration	10-55 Hz, 0.35mm

Spécifications Techniques

Normes	IEC/EN60204-1, ISOTR12100, ISO13849-1(EN954-1)
Classe de sécurité	Cat. 4 selon EN 954-1
Homologations	label CE pour toutes les directives applicables, cULus et BG
Alimentation électrique	24V c.a./c.c., 115V c.a. ou 230V c.a. 0,85 à 1,1 x tension nominale, 50/60 Hz
Consommation	2W
Contacts d'entrée de sécurité	1 ou 2 N/F ou 2 PNP barrière photoélectrique
Simultanéité des entrées	infinie
Résistance maximale d'entrée	110 ohms
Initialisation	manuelle contrôlée ou manuelle/automatique
Contacts de sortie	3 N/O de sécurité, 1 N/F auxiliaire
Puissance nominale de sortie B300 / P300 ref.: I_{th}, U_i	B300, c.a.-15, 5 A / 250 V c.a. P300, c.c.-13, 3 A / 24 V c.c.
Fusibles Sortie (externe)	6A à fusion retardée ou 10A à fusion rapide
Intensité/tension commutée min.	10mA/10V
Matière de contact	$AgSnO_2 + 0.5\mu Au$
Délai de mise sous tension	1s
Temps de réponse	15ms
Temps de rétablissement	100ms
Tension impulsionnelle admise	2500V
Indice de pollution	2
Température de service	-5 °C de à +55 °C
Humidité	90% HR
Indice de protection enceinte	IP40 (NEMA 1)
Protection aux bornes	IP20
Cablâge:	Utiliser uniquement des fils en cuivre 60/75 °C
Diamètre conducteur	0.2-4mm² (24-12AWG)
Couple des vis de bornes	0.6 Nm - 0.8Nm (5 - 7 lb•in)
Composition du boîtier	polyamide PA 6.6
Montage	rail DIN de 35 mm
Poids 24V c.c. 110V c.a. ou 230V c.a.	210g 260g
Durée de vie électrique 220V c.a./4A/880VA cosφ=0.35 220V c.a./1.7A/375VA cosφ=0.6 30V c.c./2A/60W 10V c.c./0.01A/0.1W	 100,000 d'opérations 500,000 d'opérations 1,000,000 d'opérations 2,000,000 d'opérations
Durée de vie mécanique	2,000,000 de cycles
Vibrations	10-55 Hz, 0.35mm

(g) REPAIR

If there is any malfunction or damage, no attempts should be made to repair it. The unit should be replaced before machine operation is allowed.
DO NOT DISMANTLE THE UNIT.

REPARATUR

Falls Fehlfunktionen oder Schäden auftreten, keine Versuche zur Reparatur unternehmen. Der Schalter muß ersetzt werden, bevor die Maschine wieder gestartet wird.
GERÄT DARF NIEMALS GEÖFFNEN WERDEN!

REPARATION

Dans l'éventualité d'un problème technique ou d'une détérioration de cet appareil, il doit être remplacé immédiatement avant la remise en production de la machine.
DANS TOUS LES CAS, NE DISLOQUEZ PAS L'APPAREIL.

CE Declaration of Conformity / Konformitätserklärung / Déclaration de conformité

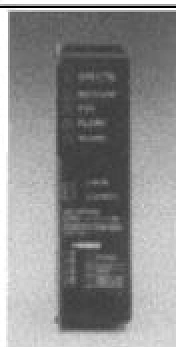
This is to declare that the Guardmaster MSR127R/T conforms with the Essential Health & Safety Requirements (EHSR's) of the European Machinery Directive (98/37/EC), the relevant requirements of the Low Voltage Directive (73/23/EEC as amended by 93/68 EEC) and the essential protection requirements of the EMC Directive (89/336/EEC as amended by 92/31 EEC). The MSR127R/T also conforms to EN 292, EN 60204-1, EN 954-1, UL 508.

Signed for Guardmaster Ltd
S. F. Mitchell
Managing Director S.F. Mitchell

Allen-Bradley
Guardmaster®

Drg No: 57516 / Issue No: 3
Change No: 0043

MP-5201
APRIL 2000



MEP100, MEP100P, MEP101, MEP102, MEP103, MEP104, MEP105,MEP230, MEP230H, MEP235, MEP236,MEP290, MEP560, MEP561 and MEP562

PROGRAMMER MODULES FOR USE WITH THE FIREYE® MODULAR MicroM™ CONTROL

Year 2000 Compliant in accordance with BSI document DISC PD2000-I:1998

DESCRIPTION

The Fireye MEP100, MEP100P, MEP101, MEP102, MEP103, MEP104, MEP230, MEP230H, MEP560, MEP561 and MEP562 Programmer Modules are used with the Fireye Modular MicroM control. The operational characteristics of the control are determined by the selection of the programmer module (e.g. re-light, 2-stage capability, pilot cutoff, etc.). The programmer module incorporates a plug-in design for easy installation.

Some programmer modules (MP230, MP230H, MP560, MP561, and MP562) are equipped with a series of dipswitches to select Purge Timing, Pilot Trial for Ignition (PTFI) Timing, Air Flow Proven, Open at Start, and Recycle or Non-Recycle operation. LED indicator lights are on all programmer modules, indicating the operating status of the control as well as providing diagnostic codes during lockout. A "check-run" switch is provided on the MEP560, MEP561 and MEP562 programmer modules to assist in testing size and stabilization of the pilot.

Flame Failure Response Time (FFRT) is determined by the selection of the amplifier module. Test jacks are also provided on the flame amplifier module to permit flame signal measurement during operation. For proper and safe application of this product, you must refer to Fireye bulletin MC-5000 for a detailed description of the various programmer modules, including installation instructions, amplifier selection, operating sequences for each programmer module, etc.

WARNING: Selection of this control for a particular application should be made by a competent professional, licensed by a state or other government agency. Inappropriate application of this product could result in an unsafe condition hazardous to life and property. Installation should not be considered complete until pilot turndown and other appropriate performance tests have been successfully completed.

PROGRAMMER MODULE SELECTION

MicroM Programmer Models:	
MEP100	Relight operation, 10 sec. PTFI.
MEP101	Relight operation, allow flame signal during "off cycle".
MEP102	Non-recycle on flame fail, 5 second PTFI.
MEP103	Fixed 10 second PTFI, 10 second MTFI, re-try once on pilot failure, post purge.
MEP104	Non-recycle on flame fail, 10 second PTFI.
MEP105	Non-recycle on flame fail, lockout on air-flow open with flame present, 10 second PTFI.
MEP100P	Relight operation, 10 sec PTFI, 15 second post purge.
MEP230	Selectable purge timing, PTFI timing, recycle/non-recycle, post purge, prove air open at start.
MEP230H	Same as MEP230 with 8 second pilot stabilization.
MEP234	Selectable air flow proven open at start, selectable purge timing, selectable PTFI, 10 second pilot proving, selectable MTFI, selectable post purge, non-recycle flame fail.
MEP235	Same as MEP230, with lockout on air-flow open 10 seconds after the start of a cycle, selectable recycle/nonrecycle lockout on air-flow open after flame is proven lockout after loss of flame.
MEP236	Same as MEP230, with additional 3 second igniter on time with main fuel. To be used with intermittent pilot only.
MEP290	Same as MEP230H, with 90 second selectable post purge.
MEP560	Same as MEP230H, 10 second main trial for ignition, run-check switch.
MEP561	Same as MEP560 without pilot stabilization.
MEP562	Same as MEP560, lockout on loss of air flow, non-recycle operation only.

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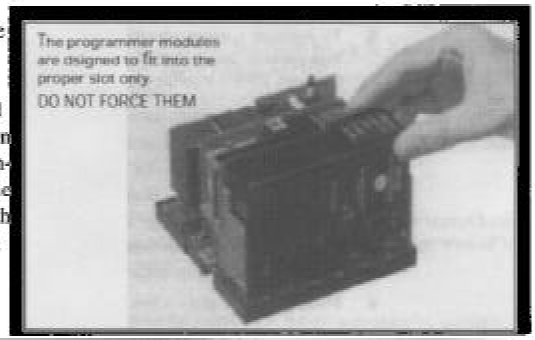
WARNING: Remove power from the control and remove the control from its wiring base before proceeding.

INSTALLATION

The Programmer Modules are used with the Fireye modular MicroM Chassis (P/N MEC120, MEC120RC, MEC120R, MEC120D and MEC120C for 120VAC and MEC230 for 230 VAC). They are installed in the chassis by grabbing hold of the programmer module by the ridged finger grips on the side on the module, aligning the module with the guide slots on the opening farthest from the transformer, and inserting the module into the pin connectors.



PROGRAMMER DISPSWITCH SETTINGS

NOTE: The dipswitch settings become permanently stored within the programmer's eeprom memory after 8 hours of continuous electrical operation. This allows sufficient opportunity to make the appropriate selection, test and checkout the system. Once stored, the settings cannot be altered.

The MEP200 and MEP500 series programmers have a series of 6 dipswitches which allow the user to program the purge timing, trial for ignition timing, enable post purge, enable proof of air flow open proven and start and select recycle/non-recycle operation.

MicroM Programmer Dip Switch Configuration

SWITCH						FUNCTION	
6	5	4	3	2	1		
				C	C	7	PURGE TIME SECONDS
				C	O	30	
				O	C	60	
				O	O	90	
			C			DISABLED	POST PURGE
			O			15 SECONDS	
		C				5	PTFI TIME
		O				10	
	C					DISABLE	PROVE AIR FLOW
	O					ENABLE	
C						RECYCLE	
O						NON-RECYCLE	

Note: C refers to switch closed position, and closed position is when the switch is toward the printed circuit board. O refers to open switch position or when the switch is moved away from the printed circuit board.

MEP200 SERIES

The MEP200 Series programmers come equipped with a bank of dipswitches that allow user selectable prepurge timing, selectable PTFI timing, selectable post purge, selectable air flow proven open at start, and selectable recycle/non-recycle operation. Refer to PROGRAMMER DIPSWITCH SETTINGS for detailed information.

Recycle operation refers to flame failure during the main (AUTO) firing period. In the event of a main flame failure, power is removed from Terminal 3 and Terminal 5. If selected by the dipswitch, the control will enter a post purge period for 15 seconds and revert back to the Idle state where the pre-purge period begins.

If non-recycle operation is selected, in the event of a main flame failure, power is removed from Terminal 3 and Terminal 5. The control will enter a forced post purge period of 15 seconds, after which the Alarm LED is illuminated and the alarm relay is energized putting power on Terminal A.

The MEP230H programmer operates the same as the MEP230 with the exception of an additional 8 second pilot stabilization. After flame is detected during the trial for ignition period, the powering of Terminal 5 is delayed for eight (8) seconds. Terminal 4 remains powered during the stabilization period. This function is offered primarily for two-stage light oil burners, to assure a specific delay between light off of the first and second stage, and to provide additional ignition timing to improve flame stabilization.

The MEP290 programmer operates the same as the MEP230 with the exception that post purge is selectable from 0 to 90 seconds.

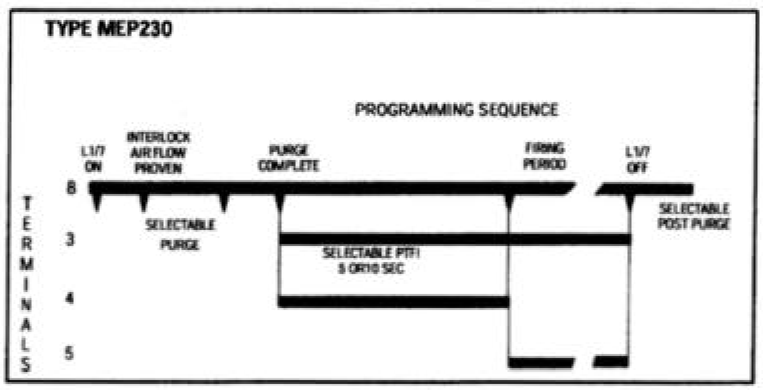
MEP235

The MEP235 programmer operates the same as the MEP230 except flame failure during the firing period causes lockout. Dipswitch #6 refers to Recycle/Non-Recycle on a loss of air flow (Terminal 6) after flame is proven. The running interlock circuit (Terminal 6) must be proven closed within 10 seconds after start of a cycle.

MEP236

The MEP236 programmer provides a 3 second main flame stabilization period by keeping Terminal #4 (igniter) energized while the main fuel valve (Terminal #5) opens. The MEP236 is to be used on an intermittent pilot only.

TIMING CHARTS



Terminal #5 is energized 3 seconds after flame is detected.
Selectable Recycle/Non-Recycle operation on loss of flame after Terminal 5 energized.
Recycle on loss of interlock (air flow) after flame proven.
Selectable air flow (interlock circuit) proven at start.

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PRODUCT GUIDE
SPECIFICATION

MINS 1185-4

FIREYE MicroM FLAME SAFEGUARD CONTROL

1. GENERAL

1.1 OVERVIEW

Each burner shall be equipped with a Burner Management Flame Safeguard Control System. The control shall provide automatic sequencing and continuous flame monitoring of the burner through prepurge (Series 200 and Series 500 programmer modules only), pilot trial for ignition (PTFI), and run.

1.1.1 The control system shall be provided by Fireye or written approved equal.

1.2 CODES AND STANDARDS

The control shall be listed by Underwriters Laboratories, Factory Mutual, and Canadian Standards Associates for its intended purposes.

2. SYSTEM HARDWARE

2.1 FLAME SAFEGUARD CONTROL

The Burner Management Flame Safeguard Control shall be of a modular design, with the programmer and amplifier modules plugging into the chassis.

2.1.1 The programmer module shall determine timing functions for prepurge (Series 200 and Series 500 models only), PTFI, and intermittent or interrupted operation.

2.1.2 The programmer module shall incorporate five LED indicator lights to indicate the operating status of the control. The LED's shall annunciate operating control closed, air flow (interlock) switch closed, pilot trial for ignition, flame signal present, and alarm.

2.1.3 The programmer module shall incorporate dip-switches (Series 200 and Series 500 programmer models only) to select purge timing, pilot trial for ignition timing, prove air-flow open at start, and recycle/non-recycle operation.

2.1.4 The programmer module shall incorporate a "run-check" switch (Series 500 programmer models only) to assist in testing the size, position, and stabilization of the pilot.

2.1.5 The amplifier module shall determine the type of flame scanner (ultra-violet, ultra-violet self-check, infra-red, flame rod/photocell, or cad cell) and flame failure response time (FFRT) of 0.8 or 3 seconds. Test jacks shall be provided on the amplifier module to measure flame signal strength during operation and shall be a uniform 0 to 10 volts.

2.1.6 The programmer module shall incorporate five LED indicator lights to indicate diagnostic status in the event of safety lockout. The cause for lockout shall be represented by a coded indication utilizing 4 of the LED's while the Alarm LED is blinking.

2.1.7 The control shall contain non-volatile memory which allows it to remember burner history and present position, even after a power interruption.

2.2 WIRING BASE

A wiring base shall be provided which will allow for all system terminations to be completely wired prior to the installation of the control. The control shall be removable or replaceable without removing any wiring terminations. The user shall have access to the wiring terminals to test for voltages while the control is installed on the wiring base.

2.2.1 The wiring base shall provide line voltage terminal inputs from direct connection of limit and operating controls, air flow switch, burner motor, ignition, pilot valves, main fuel valves, and alarm.

3. SYSTEM SOFTWARE

3.1 SEQUENCE OF OPERATION

The control shall accomplish a safe start check during each start. If flame (real or simulated) is detected prior to a start or during purge, the fuel valves shall not be energized, and the control shall initiate a safety lockout.

3.1.1 Once the operating control circuit is closed, the control shall energize the blower motor and check to prove the air flow switch is closed.

3.1.2 After the air flow switch is proven closed, the control shall initiate a prepurge (Series 200 and Series 500 programmer modules only).

3.1.3 Upon completion of prepurge, the control shall initiate pilot trial for ignition (PTFI).

3.1.4 When flame signal is proven during PTFI, the control shall de-energize the ignition transformer and energize the main fuel valve.

3.1.5 Safety shutdown shall occur following (1) failure to prove the pilot flame during PTFI, or (2) loss of flame signal after main fuel valve is energized for the selected Flame Failure Response Time and the programmer is set for non-recycle operation (Series 200 and Series 500 programmer modules only).

3.1.5 The system shall recycle automatically under control of the operating control and when power is restored following a power failure. Manual reset shall be required following any safety lockout, even after a power failure. When in a lockout condition, power interruptions will not recycle the control.

3.2 SAFETY PROVISIONS

A self diagnostic circuit within the control will identify module failures and an appropriate LED indication will be displayed for servicing. This circuit will cause a safety shutdown should any component in the control fail.

3.2.1 The control will continually test the status of all safety critical loads (ignition transformer, pilot fuel valve, main fuel valve) to insure they are operating properly.

3.3 REMOTE COMMUNICATIONS

3.3.1 The MicroM shall provide a Modbus communications protocol through the use of an optional plug-in board located on the chassis. Available information via Modbus shall be current status (lockout or run), flame signal, safety lockouts, current status, total burner cycles, burner on time, system on time. The physical configuration shall be two-wire differential RS-485 allowing up to 32 MicroM controls to be connected on each node.

3.4 REMOTE DISPLAY

3.4.1 The MicroM shall provide the capability to output message information to a remotely mounted alpha-numeric display. The messages shall be clear, concise information concerning system timing, present burner sequence position, lockout causes, historical data and control configuration.

4. PRODUCT INFORMATION

4.1 -
- The control chassis shall be the Fireye MEC120 (or appropriate model).
- The Programmer Module shall be the MEP230 (or appropriate model).
- The Amplifier Module shall be the MEUV4 ultra-violet type (or appropriate model).
- The ultra-violet scanner shall be the Fireye UV1A6 (or appropriate model).

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